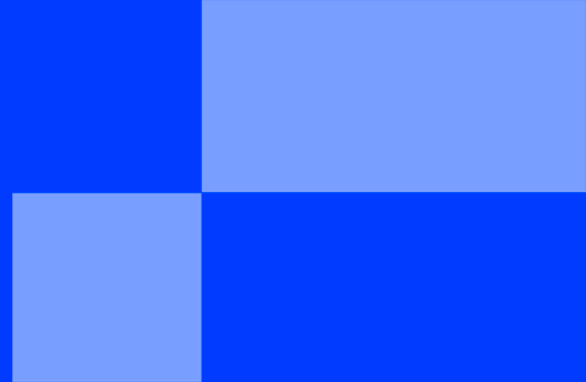




CARTEPRO



< FUNCTIONAL SPECIFICATION />

CARTEPRO

Ministère du Job et Bonheur, Digital and Innovation Directorate

Reference: JEB/DNI/2026-002

Version: 1.0, for selected technical service providers

Your contacts within the ministerial cabinet:

- ✓ Florine Pontallac, legal advisor
- ✓ Thomas Vignal, digital advisor
- ✓ Benjamin Sellami, communications advisor

1. Institutional context

As part of the policy to improve purchasing power and workplace well-being carried by the Ministère du Job et Bonheur, the **CartePro** scheme aims to modernize and extend the principle of dematerialized employee benefits.

Inspired by the meal-voucher model, CartePro offers a general-purpose tool allowing employers to credit their employees with funds usable across a network of partners referenced by the Ministry. This scheme is part of an effort to enhance the value of work and support the local economy.

2. Description of the expected service

2.1 Functional scope

The CartePro scheme relies on three distinct spaces:

Employee space:

- View available CartePro balance
- History of transactions made
- Generation of a payment QR code to present to partners
- Search and locate referenced partners

Partner space:

- Creation and management of the partner account (subject to administrative validation)
- Entry and validation of received transactions (QR code scan or manual entry)
- Financial dashboard (amounts received, transactions by period)
- Browse the catalog of other referenced partners

Administration space (Ministry):

- Validation of partner registration requests
- Account management (activation, suspension, closure)
- National dashboard (transaction volume, active partners, geographic breakdown)
- Management of employer top-ups (crediting employee accounts)

2.2 Priority functional requirements

- ✓ The balance must be displayed in real time after each transaction
- ✓ Use of the payment QR code must be able to work in degraded mode (limited connectivity)
- ✓ The partner registration process must include a manual validation step by the admin
- ✓ Transactions must be irreversible once validated

3. Technical constraints

3.1 Architecture

- ✓ Responsive web application, accessible from any mobile or desktop browser
- ✓ Backend with relational database
- ✓ Documented RESTful API (endpoints listed in the technical appendix)
- ✓ Multi-role authentication (employee / partner / admin) with session management

3.2 Transaction security

- ✓ Transaction data must be stored in an integral, non-modifiable form after validation
- ✓ QR codes must be single-use or have a limited validity period (max 5 minutes)
- ✓ Connections must be encrypted (HTTPS mandatory)

3.3 Interoperability

- ✓ The API must expose an endpoint allowing a third-party system to query an employee's balance (planned for future integration with employer HRIS systems)
- ✓ The exchange format is JSON

3.4 Performance

- ✓ Generating a payment QR code must take less than 2 seconds
- ✓ The partner catalog must support pagination for a potentially long list

4. Expected deliverables

Week 1: proof of concept (Friday project review)

- ✓ Browsable partner catalog (with at least 5 fictitious partners integrated)
- ✓ Employee space displaying balance and simulated transaction history
- ✓ Generation of a payment QR code (may be static at this stage)
- ✓ Oral presentation: functional demo + technical architecture

Week 2: final version (Thursday technical review + Friday keynote)

- ✓ All functional spaces (employee, partner, admin)
- ✓ Complete transaction process (from QR code generation to partner-side validation)
- ✓ Operational admin dashboard
- ✓ Technical documentation (installation, API, database schema)
- ✓ Project retrospective

5. Points of attention

CartePro is not a payment service in the regulatory sense within the demonstrator version developed in this context. It is a functional simulation. No real financial transaction will be carried out.

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v1

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